

David W Miller

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Senior Technical Artist

Technical Artist bridging art and engineering through shaders, optimization, and pipeline tooling

Technical Artist & Pipeline Engineer with 10+ years building production-scale pipelines and tools for AAA game development and real-time content creation. Expert in end-to-end art pipelines across multiple DCC tools and game engines. Contributed to 15+ AAA releases including Fortnite and New World, operating independently in ambiguous problem spaces while establishing pipeline standards and best practices across cross-disciplinary teams.

TECHNICAL SKILLS

3D Content Creation & Technical Art

Autodesk Maya (MEL, M.A.S.H, Bifrost. Rigging, Blendshapes, mGear, nCloth, XGen), SideFX Houdini (Environment Art & FX), ZBrush, Mudbox, Texturing (Adobe Substance 3D/Designer/Painter), Adobe Creative Suite (Photoshop, Illustrator, After Effects, Premiere)

Game Engine & Real-Time Rendering

Unreal Engine 4.x–5.x (Blueprints, C++, Python, Plugin Development, Materials/Shaders, Unreal Insights, Morph Targets, Grooms, Control Rig, LODs), Physically Based Rendering (PBR) & Stylized Materials, NVIDIA Nsight Graphics, Real-time Optimization, Motion Capture (Vicon Shogun, Optitrack Cameras)

Pipeline Engineering & Development

Python (Pipeline Tools, PyQt/PySide, PIL, FFMpeg, REST/CRUD APIs), C++, Perforce Helix Core, Git/GitHub Actions, Shotgrid, SyncSketch, Docker, Jenkins, MongoDB, AWS Deadline, Visual Studio/VSCode, PyCharm, Jira, Confluence

EXPERIENCE

Technical Director | Mar 2021 - Mar 2026

HALON Entertainment

Built custom materials and shaders, artist pipeline tools, and production automation systems supporting 20+ Unreal Engine projects spanning film, commercials and game cinematics. Developed technical art workflows for distributed teams, implemented backend asset management integrations, and provided hands-on technical support across character, environment, and sequencer-based production.

- Technical Artist on 10+ seasonal content releases for Epic Games' Fortnite, in-game intro cinematic for New World, and cinematic trailers for Tiny Tina's Wonderlands, Remnant 2, KSP2, Vampire the Masquerade: Bloodlines 2 and Borderlands 3.
- Technical Lead for ESPN SportsCenter HQ LED Wall virtual environment project, implementing Sequencer-based animation workflows and optimizing scene complexity for real-time performance within existing production infrastructure.
- Implemented vertex crowd simulation baking system for 150+ lightweight background character animations used across 3 different Fortnite season trailers.
- Built Unreal Engine to Maya MetaHuman face and body export and import tools in Python, including blendshape transfer and skin weight preservation for supporting studio animation workflow.
- Designed and implemented in-house branch of open-source AYON Unreal Engine plugin, managing distributed asset workflows, metadata synchronization, and pipeline orchestration across multiple projects and teams.
- Deployed and maintained AWS Deadline distributed render farm infrastructure, automating job scheduling, resource allocation, and failure recovery to support production at scale.
- Resolved 100+ engineering tickets spanning Python-based artist tools, system migrations, API integrations, and production bug fixes, improving pipeline efficiency and reducing turnaround time for artists and developers.
- Authored standardized PBR materials and pipeline tools used across the studio on all projects including custom layering controls and feature sets, expanding artist capabilities and saving time.

Technical Artist | Oct 2017 - Mar 2021

HALON Entertainment

- Lead Technical Artist on previsualization for Call of the Wild and other films managing skinning, rigging, establishing art workflows and providing technical support for environment, animation and lighting artists.
- Provided technical art support for numerous virtual production shoots, troubleshooting real-time rendering issues, camera tracking integration, and LED wall calibration during live capture sessions.
- Optimized Unreal Engine environments for Intel and AMD graphics card demos, implementing LOD strategies, reducing material complexity and draw calls, and tuning lighting and post-processing to hit performance targets for retail showcase deployment.
- Collaborated with asset and character artists, animators, designers, and engineers to optimize real-time assets, integrate complex scenes, and build custom materials while maintaining high visual fidelity under production deadlines.

Engine Artist & Game Developer | Feb 2017 - Aug 2017

MatterVR

- Shipped mobile VR game Wonders of the World for GearVR using early Unreal Engine 4, handling full technical art pipeline from asset integration to performance optimization.
- Built custom VR rendering pipeline, profiling and optimizing materials, meshes, and lighting to maintain consistent 60fps targets across mobile hardware constraints.
- Developed real-time multiplayer minigames in BP & C++, implementing networked gameplay for historical board games with data-driven progression systems.
- Created modular material systems and reusable art components, establishing technical standards that improved asset creation efficiency across the project.

JSC Distance Learning Instructional Designer | July 2014 - Oct 2016

NASA Johnson Space Center, GHG Corporation

- Delivered +200 interactive online lessons to the Flight Operations Directorate software familiarization and classroom-based instruction, providing refresher briefings and training courses to flight controllers, instructors and astronauts.
- Developed customized training overviews for ISS laptop software by meeting with stakeholders and developers and hosting planning sessions.
- SharePoint developer & administrator, supporting instructors and division pages across the Training Standards & Integration Production Group.
- Prototyped a fully-functional ISS Environment in Unreal Engine 4 for Interactive Emergency Training.

JSC Distance Learning Training Standards Intern | Sep 2011 - Jul 2014

NASA Johnson Space Center, United Space Alliance/LZ Technology

- Built 3 ISS Emergency Response online training courses using Articulate Storyline for Fire, Depress or Toxic Atmosphere scenarios using training footage from station mockups.
- Camera Operator for Video Lab capture of Flight Controller and Astronaut training footage and briefings.

EDUCATION

M.A. Digital Media Studies / University of Houston | Dec 2016

Compositing, Illustration, 3D Modelling & Animation

B.A. Communications / University of Houston | May 2011

TV Production Studio, School Newspaper & Magazine